

7.05 Decision Making in Project Management

7.05a	Critical path analysis	a) Be able to construct and interpret activity networks using activity on arc. <i>Appreciate that a path of critical activities (a critical path) is a longest path in a directed network.</i>	
7.05b		b) Be able to carry out a forward pass to determine earliest start times and find the minimum project completion time, and to carry out a backward pass to determine latest finish times and find the critical activities. <i>Includes understanding and using the terms "burst" and "merge".</i>	
7.05c		c) Understand, and be able to calculate, (total) float.	d) Be able to find latest start times and earliest finish times for activities. Calculate and interpret independent and interfering float.
7.05d			e) Be able to use an activity network to construct a cascade chart and use a cascade chart to determine the effect on the minimum project completion time of a delay to one or more activities, or other scheduling restrictions.
7.05e			<i>Cascade charts may be constructed either with one activity on each row or with the critical activities together in one row and a row for each non-critical activity.</i> <i>Float may be shown using dashed lines.</i> <i>Includes constructing a schedule to show how a given number of workers can complete a project subject to given constraints.</i>